



NITI Aayog

DRAFT



SECTOR AT A GLANCE

Textiles

Manufacturing in India

DRAFT

Sector at a Glance

Textiles

Manufacturing

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). <Report Title>. NITI Aayog <Sector-at-a-Glance> Series – <Sector Name> in India, <Product Number>, <link>

Copyright © 2023, NITI Aayog, Government of India

All rights reserved. This draft document and the contents therein are internal to NITI Aayog. They cannot be reproduced in any manner either in full or parts, thereof. The draft document is subject to further refinements.

All information, ideas, views, opinions, estimates, suggestions, and recommendations (hereinafter 'content') in this "portal resource" should neither be understood as professional advice in any manner nor interpreted as policies, objectives, opinions, or suggestions of NITI Aayog. Readers are advised to use their discretion and seek professional advice before taking any action or decision based on these contents. The contents in this portal have been obtained or derived from reliable sources as far as possible. But, NITI Aayog neither represents this information to be accurate nor assumes any liability for any loss, or damages, caused due to any reason whatsoever, towards its usage by a third party. Readers of this document are made aware that the document may be subject to revisions. Any suggestions/input may please be sent to nitiforstates@gov.in.

www.niti.gov.in

Overview



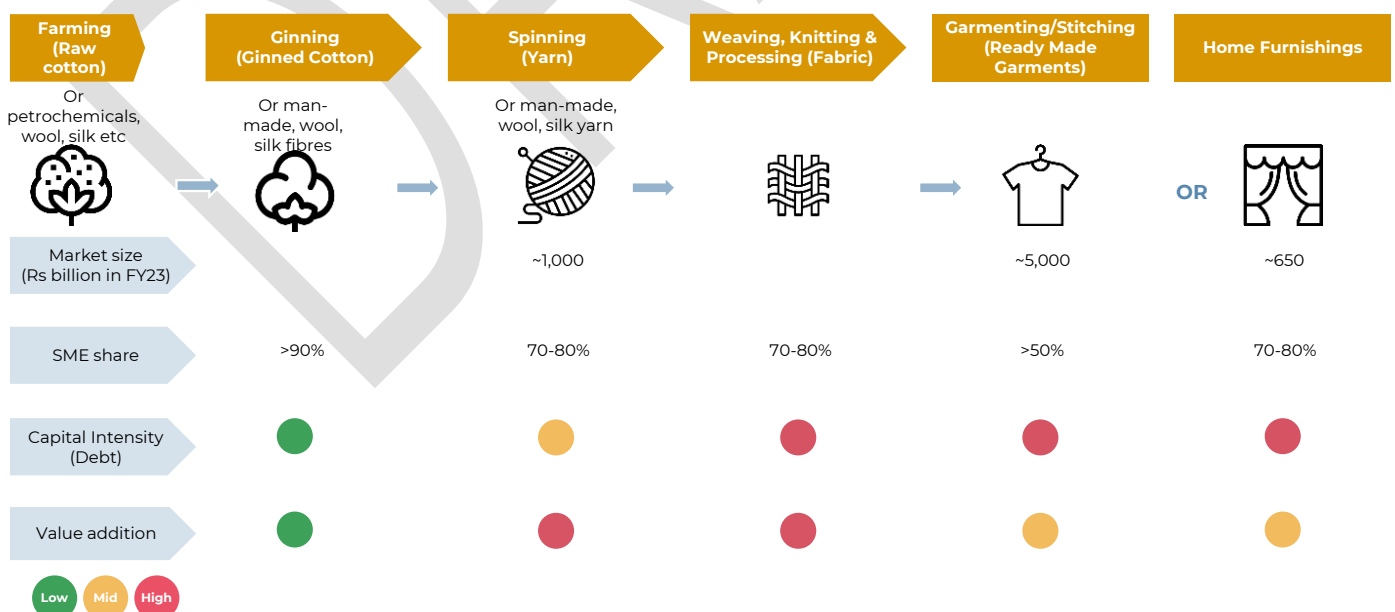
The textile sector contributes ~2%¹ to overall gross value added in India. The sector employs 45 million people. Small and medium-sized enterprises (SMEs) account for ~70% (CRISIL MI&A Research) of the total production. The industry includes segments such as cotton and synthetic yarn, cloth and readymade garments (RMG). The value chain encompasses the entire life cycle of textiles starting from the cotton crop stage to the end-use segments of RMG and home furnishings.

The textile industry plays an important role in India's economic growth on account of its share in employment, value addition and export earnings. The industry is one of India's largest employers, especially in the unorganised and non-agrarian segment and contributes ~2%¹ to overall gross value added. The number of employees working in the textile industry stood at close to 45 million², according to the Ministry of Textiles. Traditional sectors such as handloom, handicrafts and power looms are the largest sources of employment for people in rural and semi-urban areas. The industry is a major source of livelihood for a large number of women too.

The industry includes segments such as cotton and synthetic yarn, cloth and readymade garments (RMG). The value chain encompasses the entire life cycle of textiles starting from the cotton crop stage to the end-use segments of RMG and home furnishings.

The industry is also highly fragmented, with non-integrated production of small and medium-sized enterprises (SMEs) accounting for ~70% of the total production. These SMEs are concentrated in clusters across India, specialising in segments throughout the value chain.

Figure 1: Industry structure



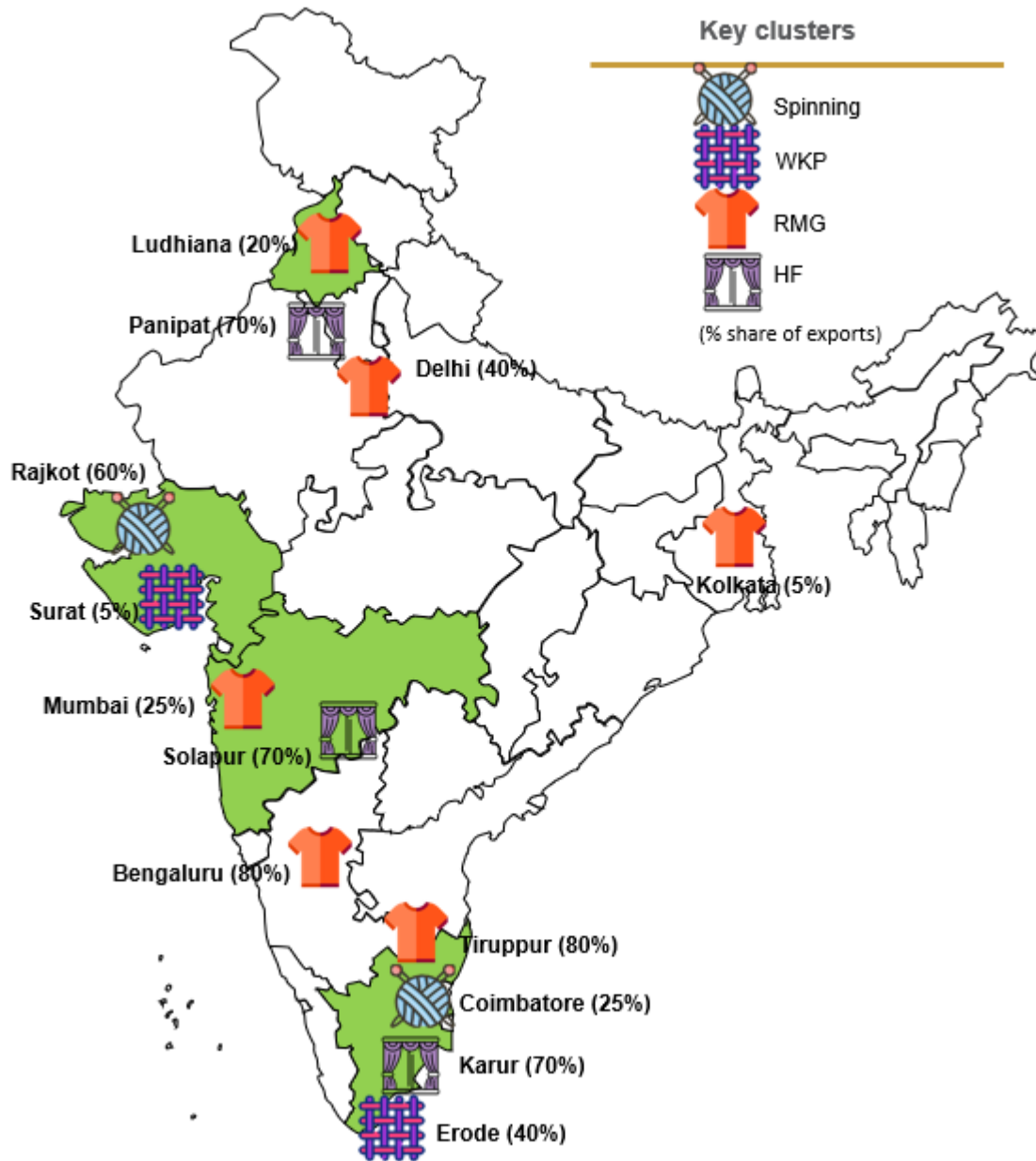
Note: Includes cotton and man-made fibre (MMF) value chain from the spinning level as the value chain starts from farming. The MMF value chain is covered in the MMF yarn section. Source: CRISIL MI&A Research

¹ National Statistical Office, Government of India

The cotton cycle in India begins with sowing in March, April and May, followed by harvesting, which is done in September and October, after which, freshly harvested cotton is sent to ginneries. Ginned cotton is then spun into yarn, while MMF is produced from crude-linked derivatives. Spun yarn is then woven and knitted into fabrics which are then used as an input for production of readymade garments and home textiles.

Spinning is a highly capital-intensive process, requiring significant investments in plant and machinery. With the continued availability of fiscal incentives, entities in this segment usually operate with relatively high levels of leverage, which results in interest and repayment burden. In such a scenario, the ability to consistently sustain high capacity utilisation levels is of utmost importance. In comparison, the weaving, RMG and home furnishing segments are less capital-intensive.

Major textile clusters



Source: CRISIL MI&A Research

Textile production in India is cluster-oriented and concentrated in specific states. For example, cotton spinning primarily takes place in the Rajkot cluster in Gujarat and the Coimbatore cluster in Tamil Nadu, as does weaving, knitting and processing in Surat and Erode located in the same states. Garment production takes place in Tiruppur (Tamil Nadu), Bengaluru (Karnataka), Mumbai (Maharashtra), Kolkata (West Bengal), Delhi and Ludhiana (Punjab). The reliance on export and domestic markets varies across clusters. As a result, moderation of export demand has a larger impact on clusters such as Tiruppur, where the exposure to export markets is high compared with the Mumbai or Kolkata market, which relies on domestic demand.

Sub-sectoral trends

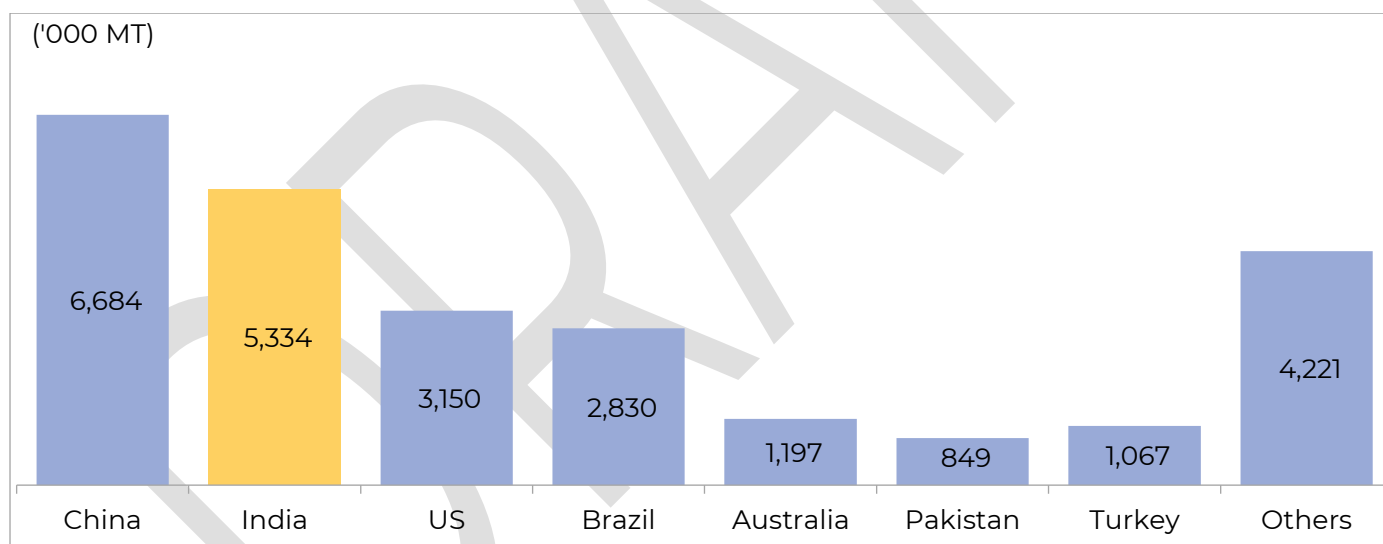
1. Cotton

India is the second-largest cotton producer in the world, accounting for over a fifth of global cotton production as of cotton season 2022/23

India also has the highest area under cotton cultivation among all cotton growing countries

Cotton is one of the major crops of the country and is a key raw material for the domestic textile industry. Therefore, area under cotton cultivation in India is the highest among all cotton growing countries, according to the US Department of Agriculture. Its vast cotton acreage enables India to make a sizeable contribution to global production. India is the second-largest cotton producer in the world, trailing China, accounting for 23%² of global production on average during CS2020-23. Global cotton season commences from August.

Figure 3: Global cotton production (cotton season 2022/23)



Note: The information represents the global cotton season commencing from August 1, 2022 and ending on July 31, 2023

Source: US Department of Agriculture

In India, a majority of cotton is produced in Gujarat, Maharashtra and Telangana. However, across the country, cotton is cultivated in three varied agro-ecological zones – the northern zone comprising states such as Punjab, Rajasthan and Haryana, the Southern zone made up of Telangana, Tamil Nadu, Andhra Pradesh and Karnataka. The central zone comprises Gujarat, Maharashtra, Madhya Pradesh and Odisha. A majority

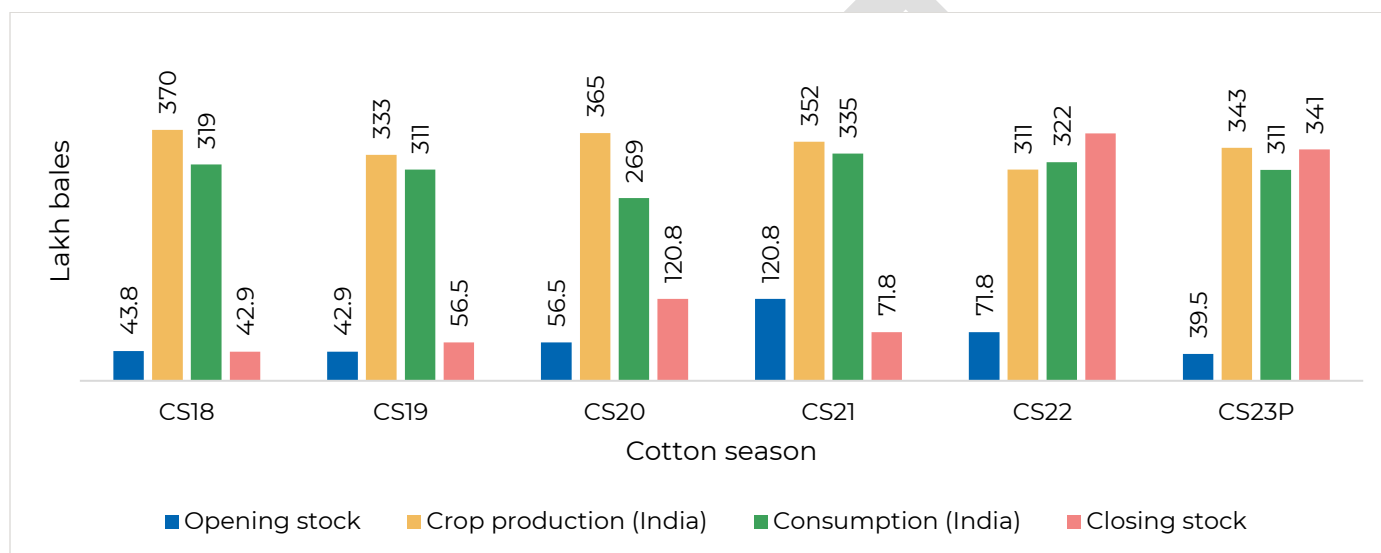
² Ministry of Textiles

of cotton grown in India is produced on rain-fed areas and close to 40%³ is grown on irrigated land, according to the Ministry of Textiles.

4

Indian cotton is of medium and long staple varieties. Medium staple cotton, of 26 mm to 28 mm length, has stronger and low-maintenance fibres. Long staple cotton, of 29 mm to 34 mm length, is used for producing high-quality products, given its premium and smoother feel. In India, raw material consumption is skewed towards cotton over MMF and filament yarn in the ratio of 59:41³.

Figure 4: India's cotton production trend



Note: The cotton season in India represents the period from October to September

Source: Cotton Corporation of India

Apart from usage by the downstream textile industry, cotton produced in India is also exported mainly to China, Bangladesh, Vietnam, and other Southeast Asian countries. Of these, Bangladesh and China are the largest importers of cotton from India. At present, cotton exports from India do not have any restrictions in terms of permissible quantities that can be supplied. Given India's pole position in cotton production, consumption by the downstream industry is primarily met by domestically grown cotton. However, India does import the extra-long staple variety of cotton, which is not grown domestically.

Cotton yarn

It dominates the fibre mix in domestic consumption

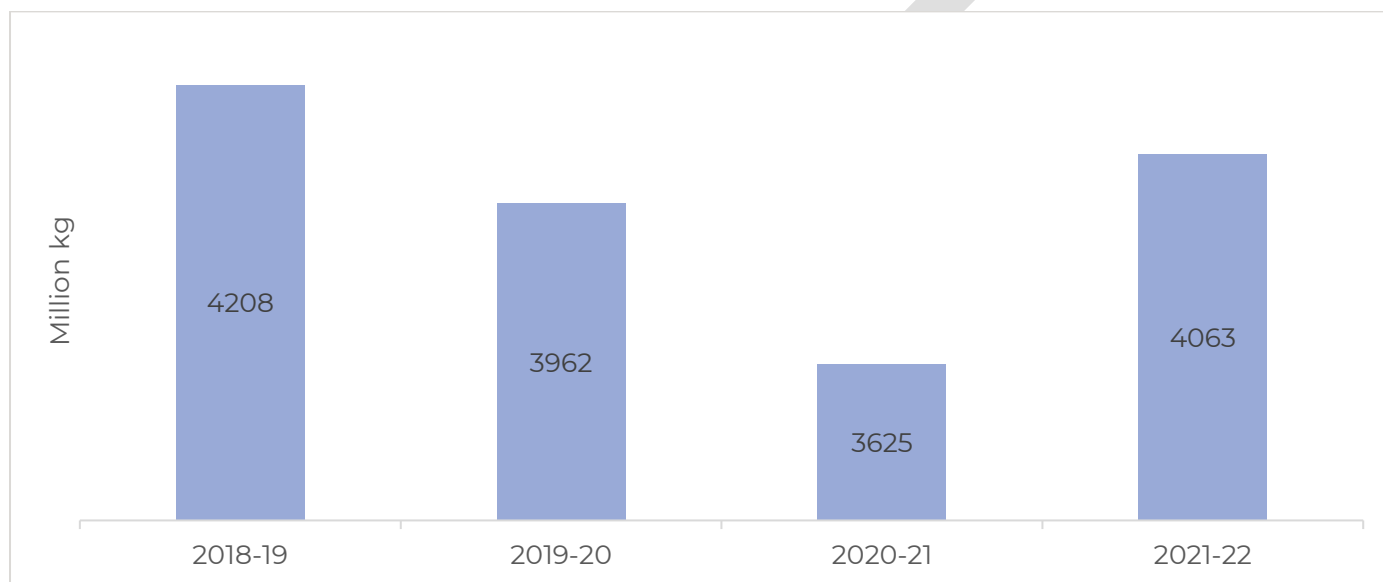
Cotton spinning is the process of producing yarn from cotton fibre. Yarn which does not require spinning for production is called filament yarn, while yarn produced through the

³ Ministry of Textiles

process of spinning is called spun yarn. In India, the share of cotton yarn in overall consumption basket is higher than that of synthetic yarn. As against this, global consumption is skewed towards synthetic yarn.

The Indian cotton yarn industry is vibrant and benefits from availability of domestically produced cotton. As apparel manufacturing moved from high-cost destinations in Europe to low-cost geographies in Asia, self-reliance enabled India to draw strength from its unique backward linkages with cotton, unlike most peers.

Figure 4: Cotton yarn production trend



Source: Office of the Textile Commissioner

Most of the yarn produced is used by the domestic downstream RMG industry and close to a third is exported. In India, yarn is produced by both large players having state-of-the-art technology and machines and smaller companies in the SME segment. The fibre length of yarn produced in India usually ranges from 30 to 40 mm⁵

India's cotton yarn consumption is driven by steady demand for RMG, which in turn is driven by rising discretionary income, increased urbanisation, the festive and wedding season, workwear, womenswear, and kids wear and of late, athleisure.

India the largest cotton yarn supplier in the world

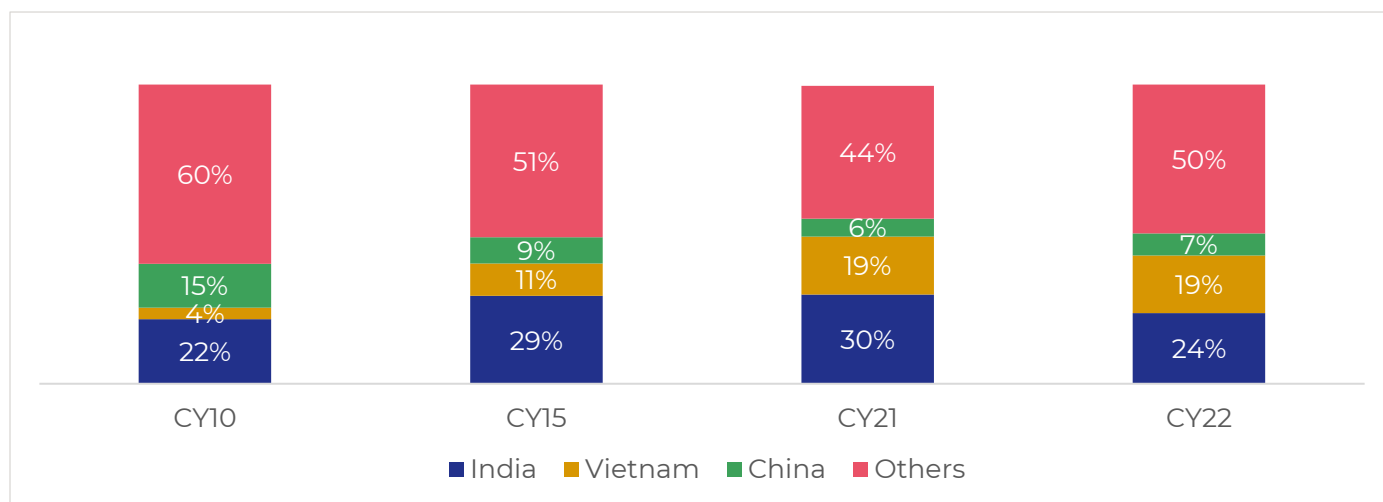
Being one of the largest cotton producers in the world, India has a significant share in the global export market of upstream textile products such as cotton and cotton yarn. Based on ITC Trade map, the global market primarily has three suppliers of cotton yarn – India, China, and Vietnam. Together, these three countries account for close to half of cotton yarn exported every year.

As a result, the country is a major raw material supplier to textile units in Southeast Asia. Cotton yarn from India is mainly exported to China, Bangladesh, Vietnam, among

⁵ Ministry of Textiles

others. These countries have a presence in the RMG segment and compete with India in global RMG trade.

Figure 5: Share of exporters in cotton yarn trade



Source: ITC Trade Map

India continues to be the largest exporter of cotton yarn in the world primarily on account of strong backward linkages with domestically produced cotton. Data from ITC Trade Map over the past decade shows that India consistently accounts for over a fifth of cotton yarn exported. China, which made up around 15%⁴ of global cotton yarn exports in 2010, has seen its share shrinking to ~7% in recent years. China's share in global cotton yarn supply has been on a declining trend following the ban on Xinjiang cotton, which makes up most of the cotton grown in the country, as well as intermittent lockdowns in China which impacted economic activities. This indicates further potential for Indian cotton yarn in the global market.

2. Man-made fibres

Types of man-made fibres

The landscape of global textile production and trade is chiefly shaped by two prominent categories of fibre-based products: cotton and man-made fibre (MMF). Historically, the textile and apparel (T&A) production chain has been dominated by natural fibres such as cotton. However, in recent years, there has been a remarkable shift in the T&A industry, with MMF-based products emerging as the dominant force in both global production and trade.

MMFs can be broadly categorised into two types: synthetic and cellulosic.

Synthetic fibres are produced from raw materials, primarily sourced from petroleum derivatives, and rooted in the realm of chemicals and petrochemicals. These materials undergo polymerisation, which results in the formation of long, linear chemical

compounds. These compounds serve as the foundation for the creation of diverse types of fibres, each imbued with unique characteristics.

4 ITC Trade Map

In contrast, cellulose fibres are derived from ethers or esters of cellulose, a compound that can be extracted from bark, wood, leaves or other plant-based materials. The production of cellulose fibres begins with the processing of specific plant sources into pulp. Among the most prevalent 'manufactured' cellulose fibres, viscose is a widely used textile derived from rayon-grade wood pulp (RGWP).

Table 2: Indian MMF industry

Classification of the Man-Made Fibers			
Synthetic		Cellulosic	
Polyester	75-77%	Viscose	16-18%
Nylon	4-6%		
Acrylic	0-2%		
Polypropylene	0-1%		
Key products and Share in consumption mix			

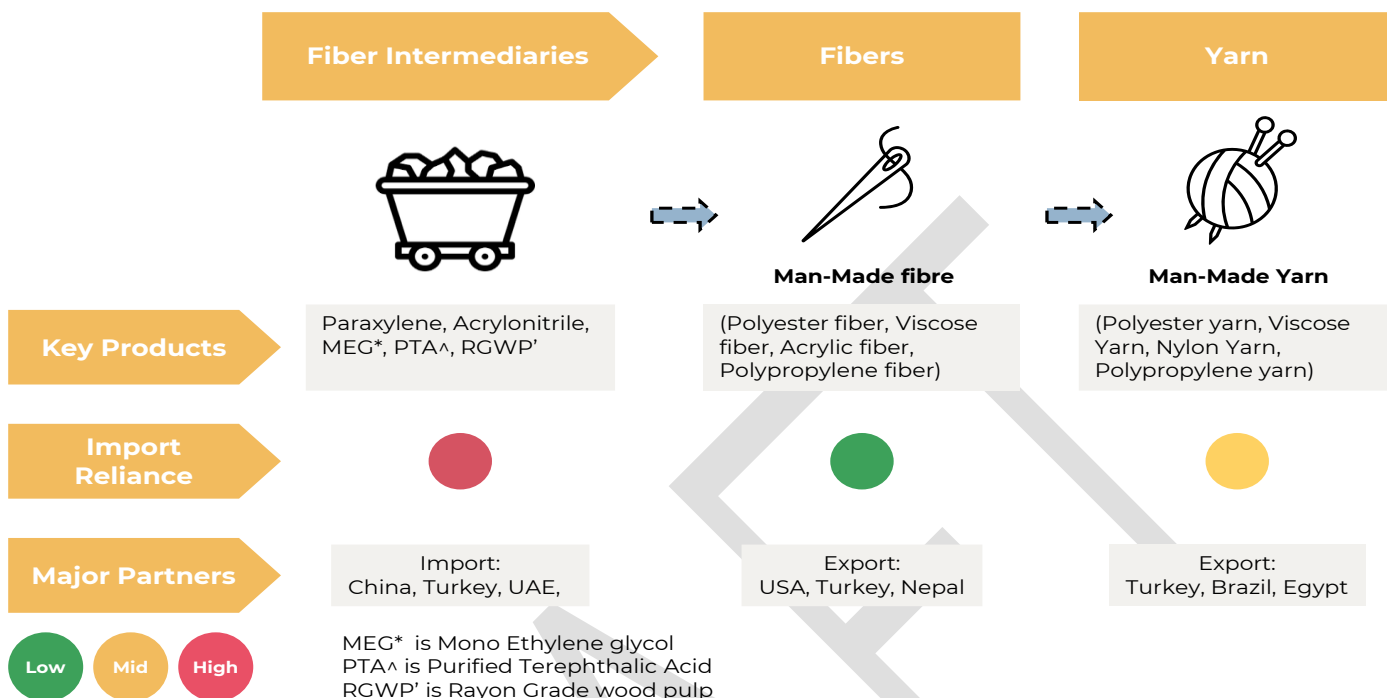
Note: Computation basis aggregating data of MMF and yarn

Source: Chemicals & Petrochemicals Statistics at a Glance - 2022, CRISIL MI&A Research, various company reports

Understanding MMF value chain

The MMF value chain begins with the initial stage of raw material production, followed by creation of intermediate products including fibres and yarns. These intermediates are then either used independently or blended with natural fibres. Ultimately, this process leads to the production of final consumer goods such as fabrics and ready-made garments.

Figure 6: MMF value chain



Source: Chemicals & Petrochemicals Statistics at a Glance - 2022, CRISIL MI&A Research, various company reports

Fibre intermediaries: India is among the world's largest intermediary manufacturers for key raw materials such as purified terephthalic acid (PTA) and monoethylene glycol (MEG). Despite this strength, rapid expansion in both midstream and downstream segments has led to a significant reliance on imports for key raw materials. That said, there is a notable shift underway within Indian refineries, moving towards greater petrochemical integration. This shift is anticipated to reduce import dependence in the medium term. To achieve self-reliance, there is a pressing need for incremental capacity expansions. As these products heavily depend on crude derivatives, it is essential to highlight that, according to data released by the Ministry of Petroleum and Natural Gas (MOPNG), India currently imports a substantial 85-90%⁶ of its crude oil needs. This underscores the importance of closely monitoring India's competitiveness with global counterparts.

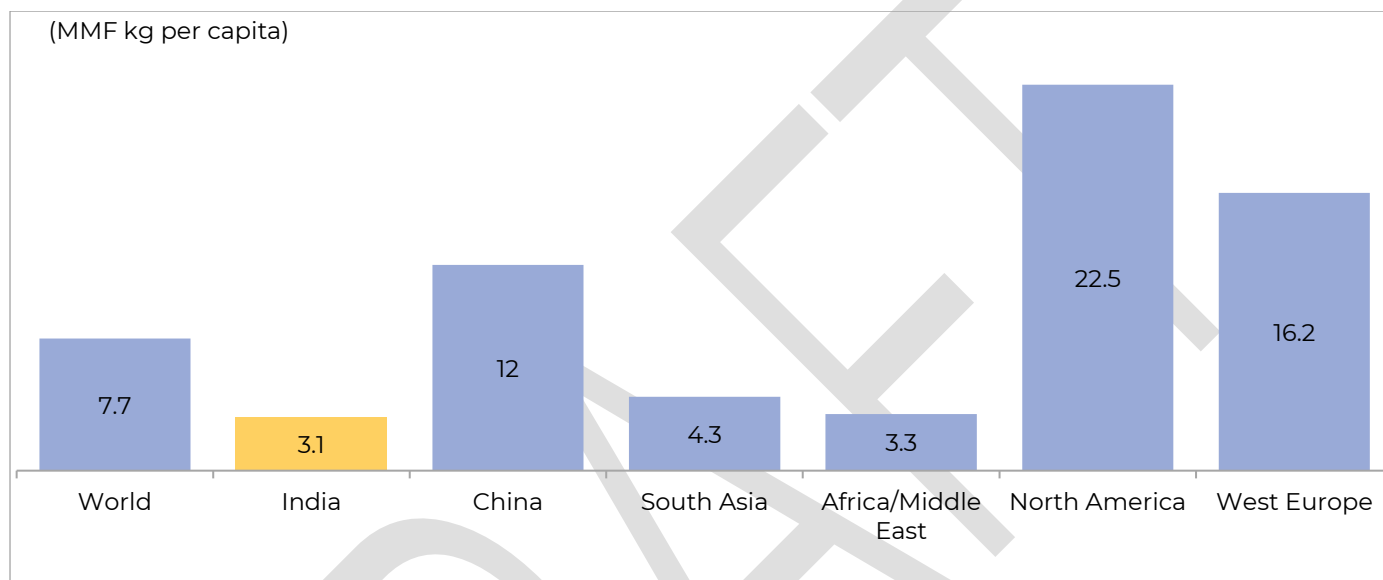
Fibre and yarn: High demand for textiles, coupled with a substantial textile manufacturing ecosystem, has positioned India as a net exporter of specific MMFs and yarns. Polyester, as the most extensively consumed product, has seen remarkable capacity expansion in India, largely supported by stable domestic demand owing to its role as a viable substitute for natural fibres. Meanwhile, the viscose market — the second-largest segment within the MMF industry — is experiencing rapid growth. However, limited capacity for viscose yarn production has resulted in a substantial reliance on imports to meet domestic demand.

⁶ Basis data compiled from PPAC (Petroleum planning and Analysis cell)

Indian MMF-yarn consumption story: Long road ahead

India is poised to emerge as one of the world's fastest-growing large economies, marked by substantial improvements in GDP and per capita income. This economic upswing is expected to lead to a notable increase in discretionary spending, thereby fostering a steady rise in per capita textile consumption.

Figure 7: Global per capita MMF consumption



Source: Ministry of Textiles

MMFs are expected to account for a significant share (70%) of the global T&A market. In stark contrast, India's T&A industry has historically favoured natural fibres (a 30:70 ratio), especially cotton, over MMFs.

However, this landscape is expected to undergo a transformative shift. The growing preference for blended textiles, coupled with the rising per capita income of the population, is expected to drive substantial growth in MMF consumption within India. This trend represents a pivotal development, aligning India's textile market more closely with the global landscape and reflecting evolving consumer choices and economic progress.

3. Readymade garments

India: Global RMG manufacturing hub

The process of garment manufacturing typically starts with the design of the product, which begins about three to four seasons in advance, depending on forecasts of fashion trends. After finalising design features such as cloth, colour, styling, and quantities, vendors engage in sampling and costing processes.

The global RMG market is price-competitive, and supply mainly comes from Southeast Asian countries where labour is available abundantly and at competitive rates. Raw material is easily accessible, which helps lower manufacturing costs and keeps prices of finished goods low.

The domestic RMG industry has low reliance on imports on account of self-sufficiency in terms of natural as well as synthetic fibre availability. RMG production is highly unorganised in India, fragmented and spread across different clusters having varied levels of exposure to the domestic and export markets. Close to a third of garments manufactured are exported, while the rest are consumed in India.

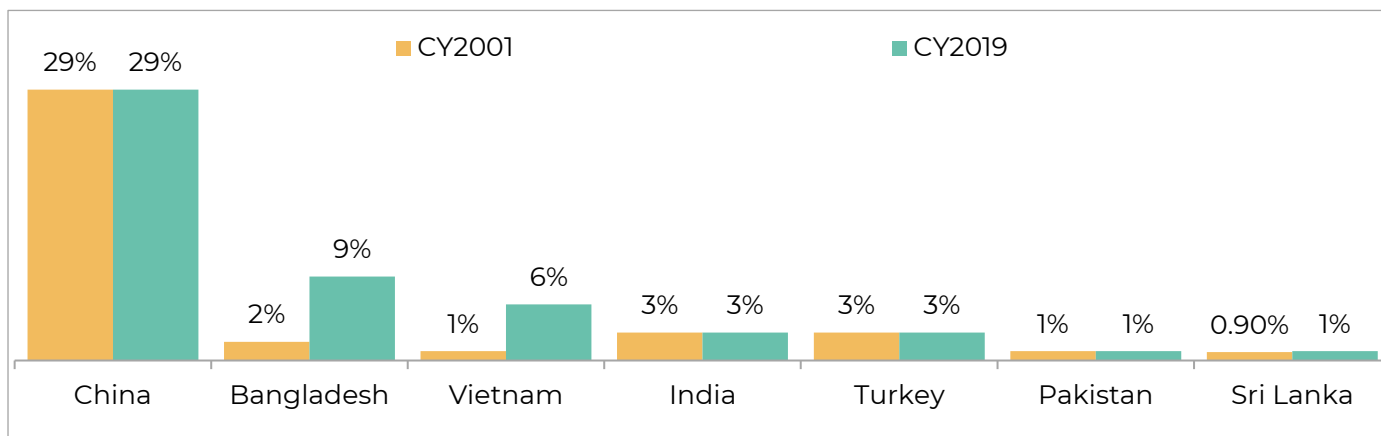
India key RMG exporter, but growth in share restricted

India has an edge over other RMG suppliers on account of domestically available cotton. The country competes with China, Bangladesh, and Vietnam in global RMG trade. China has traditionally been a dominant player in RMG trade globally. The country has large-scale manufacturing facilities throughout the segment and offers complete products to customers at a competitive price. Over the past decade, Bangladesh and Vietnam have also registered consistent growth in apparel trade.

Developed economies such as the US, the EU and the UK are the leading apparel consumers. As apparel manufacturing is a labour-intensive process, these countries rely on Asian countries for RMG supply. As a result, there is a distinct separation of the textile value chain into production hubs and consumption.

The share of the US and the EU in RMG exports by India, China, Bangladesh, and Vietnam is huge. As a result, the RMG industry, as well as the upstream textile industry, is vulnerable to discretionary demand emerging from these countries. Both these regions account for more than half of garments exported by India.

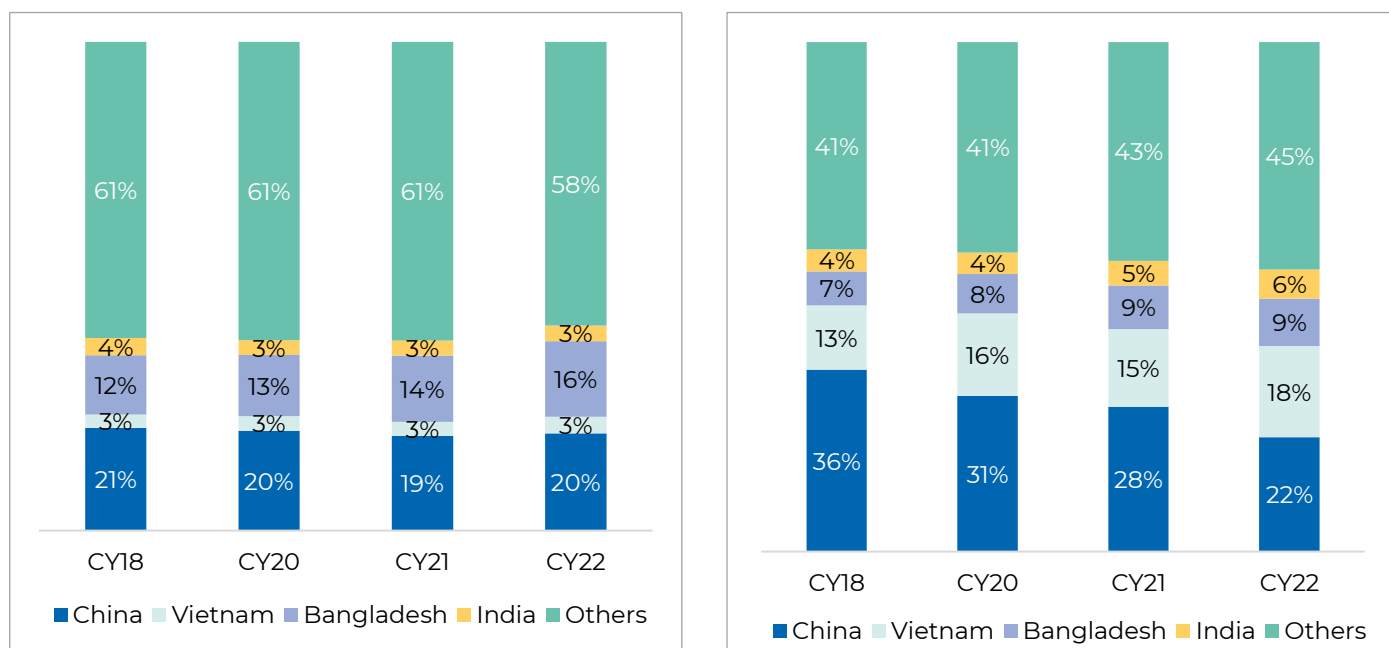
Figure 8: Country-wise trend in RMG exports



Source: ITC Trade Map

The chart above signifies the positioning of India and other leading RMG exporters in the global market and their share over the past two decades. China continues to hold the largest share in apparel trade and has consistently accounted for almost a third of RMG exports in the world. Bangladesh and Vietnam, due to their competitive labour costs and technologically upgraded machinery, as well as proximity to China, have managed to increase their contribution to exports sizeably. In contrast, India's share has remained stagnant at 3% over the past two decades due to relatively high labour costs, skill disparity, lack of technological upgradation translating into lower efficiency, and absence of trade pacts.

Figure 8: Share of exporters in RMG imports by the EU (LHS) and the US (RHS)



Source: ITC Trade Map, CRISIL MI&A Research

While India's RMG exports are driven by the EU and the US, India's share in total imports by these countries remains very low — 3% and 6%, respectively.

Moreover, India's share has remained stagnant at these levels over the past five years. And this is in spite of India benefitting from ample raw material availability. In comparison, competitors such as Bangladesh and Vietnam have a higher and increasing contribution to US and EU imports.

China continues to be the largest supplier of RMGs to the US and the EU, accounting for around 20% of RMG imports by the EU and the US over 2018-23. Vietnam's share in US imports has risen by five percentage points over the course of just four years. Bangladesh has also registered growth, albeit at a lower rate. In EU imports, Vietnam has a much lower contribution of 3%. Bangladesh's share in imports by the EU jumped from 12% to 16% within just four years.

Challenges

1. Duty differential

Indian textile exports face higher duties in comparison with competing nations such as Bangladesh, Pakistan and Vietnam.

India attracts higher duties than Vietnam and Pakistan on cotton yarn exports to China, a major RMG producer. While India's cotton yarn exports attract a 3.5% duty, Vietnam and Pakistan have duty-free access to Chinese markets, which helps improve their competitiveness.

Table 2: Duty differential — cotton yarn

	Importers			
	China	Vietnam	Bangladesh	Pakistan
Exporters				
India	3.50%	5%	3%	5%
Vietnam	0%	-	10%	11%
Pakistan	0%	5%	3%	-
China		5%	10%	5%
Others	5%	5%	10%	11%

Source: ITC Market Access Map

India faces a disadvantage compared with Pakistan, Bangladesh and Vietnam in exporting RMGs to the EU and Canada. India's RMG exports attract a 9.6% duty when shipped to the EU and 18% when shipped to Canada. However, Vietnam and Bangladesh do not attract any duties while exporting garments to the EU and Canada. In the case of Pakistan, exports to the EU attract zero duty. Considering India competes in the lower-priced garments segment, a comparatively high duty affects the attractiveness of Indian textiles.

Table 3: Duty differential – RMG

	Importers			
	EU	US	Canada	China
Exporters				
India	9.60%	10.15%	18%	5.60%
China	12%	10.15%	18%	-
Vietnam	0%	10.15%	0%	0%
Bangladesh	0%	10.15%	0%	0%
Pakistan	0%	10.15%	18%	7%
Others	12%	10.15%	18%	6%

Source: ITC Market Access Map

2. Higher labour costs

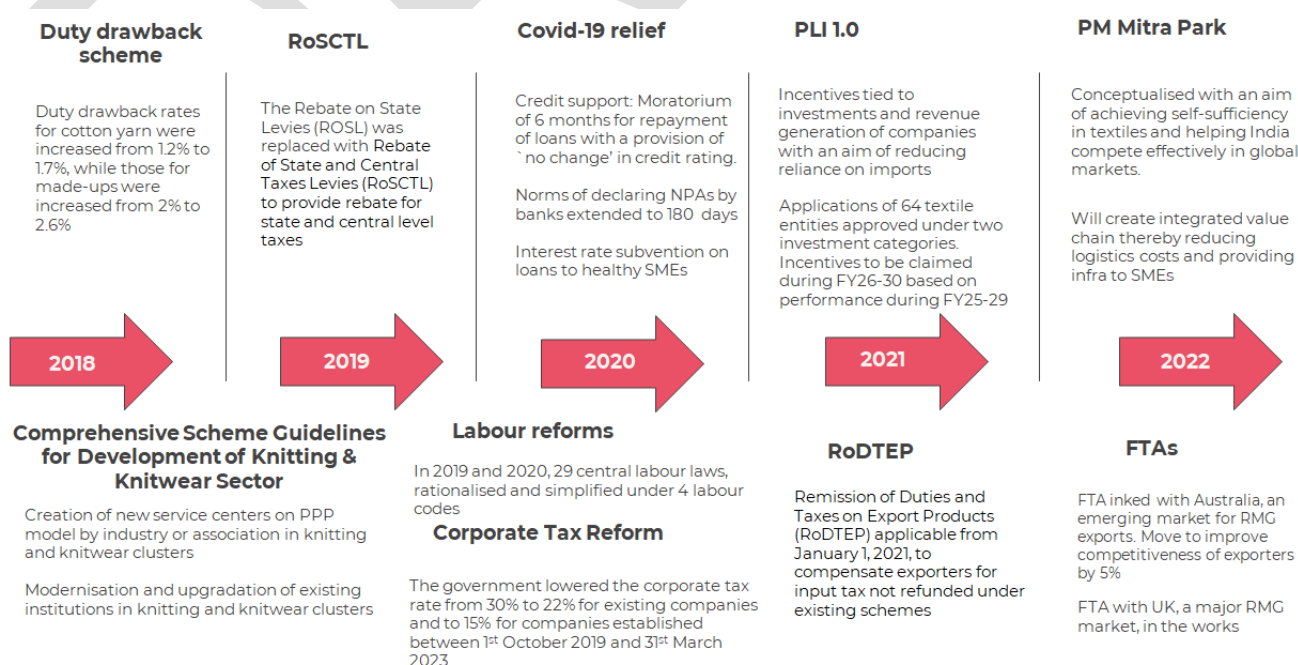
India's labour costs, while lower than those of Vietnam and China, are significantly higher than those of Bangladesh.

3. Low productivity

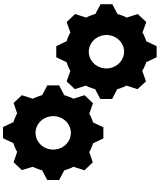
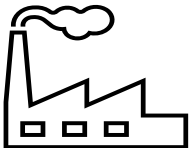
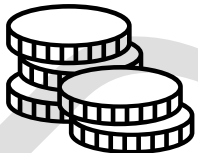
Vietnam and China have technologically upgraded machinery, resulting in higher productivity levels compared with India.

Policy tailwinds

Evolution of policy vision addressing areas of improvement across the value chain to spell faster growth



Policy tailwinds (Continued)

	PLI Textiles is one of the 13 industries included under the Production-Linked Incentive (PLI) scheme announced in Union Budget 2021-22. The scheme aims to generate ~2.5 lakh jobs in the textile sector and its supporting activities; promote production of fabrics based on high-value man-made fibre (MMF), RMG and technical textiles; and encourage introduction of new capacities and address demand for high-value MMF textiles in the country by generating more employment and increasing trade.
	PM MITRA Under the PM MITRA Parks scheme, the government plans to set up seven mega textile parks over seven years up to fiscal 2027, with an outlay of Rs 4,445 crore. The government intends to create 1 lakh jobs directly and 2 lakh indirectly, per park, through the scheme. The scheme aims at creating an integrated value chain from spinning to garment manufacturing at a single location, to help reduce logistics costs, while making facilities accessible to smaller entities.
	RoDTEP and RoSCTL The government announced the Remission of Duties or Taxes on Export Products (RoDTEP) and Rebate of State and Central Taxes and Levies (RoSCTL) schemes to neutralise taxes and duties on exported goods that are otherwise not credited, remitted or refunded. This includes both direct costs incurred by the exporter and cumulative indirect taxes at prior stages. The schemes help exporters meet international standards, while rendering products cost-competitive in the international market, and also offer a stable policy regime, while levelling the playing field for exporters.
	FTAs Lack of free trade agreements (FTAs) with key markets puts India at a disadvantage compared with competitors, especially in the price-competitive apparel segment. Signing FTAs with developed nations that are major consumers of RMGs will help

	enhance India's position in the global market and support the sustainable growth of India's RMG industry.
--	---

Way forward

Demand for Indian textiles is expected to log significant growth over the next decade, driven by sustained improvement in **domestic demand** and increased **competitiveness** of India's textile industry in the global market. Rising discretionary spending, expanding middle class, brand awareness among consumers, and demand for workwear and athleisure are likely to contribute to domestic demand. Demand will also be supported by steady improvement in export markets, while initiatives such as export-friendly incentives and **FTAs** will boost the competitiveness of Indian textiles in the global market.

Demand

India is likely to retain its **pole position** as the **leading supplier** of cotton yarn, backed by abundant domestic availability, which in turn will benefit from geographical advantage, large area under cultivation and support from the government in terms of the **minimum support price (MSP)** for the crop. Yarn exports are poised to gain from steady growth in underlying RMG demand, which will result in higher offtake of Indian yarn by other RMG producers. The global **shift in sourcing away from China** is likely to continue to offer support. Additionally, **trade pacts** with key markets will offer scope for improvement in market share. India also stands to gain from initiatives such as **PLI** and **PM MITRA Parks**.

Supply



सत्यमेव जयते

NITI Aayog